Labour Market Power in Canada: Long-Term Trends and Cyclical Dynamics

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Abstract

This paper provides the first time-series analysis of labour market concentration in Canada. Using linked employer–employee data for 2001–2019 and a wage-bill Herfindahl index, I show that national and local concentration declined over the long run but diverged during the Great Recession, when aggregate local concentration rose by 14 percent. Economy-wide, Public Administration accounts for 54 percent of the long-run within-province decline in local concentration, consistent with the federal restructuring that followed Budget 2012, and wage-bill reallocation toward this highly concentrated sector explains 68 percent of the within-province rise during the recession. In the private sector, the long-run decline comes from falling concentration within industries, reinforced by a shift of the wage bill toward less concentrated industries; net firm entry offsets about a quarter of the decline generated by continuing firms. During the recession, private-sector concentration rose mainly within industries, and entry and exit amplified the increase. With a leave-one-out instrument, the market-level firm wage premium falls with concentration, with an elasticity of -0.028. Finally, a two-agent New Keynesian model with an endogenous wage markdown shows that government spending narrows markdowns and shifts income from profits to wages, more strongly than in a model without cyclical monopsony.

JEL Codes: E32, E62, J31, J42, L16.

Keywords: Labour market concentration, monopsony power, Great Recession, public sector, fiscal policy.

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1 Introduction

This paper studies the evolution of employer power in Canadian labour markets and how the Great Recession reshaped the competitive structure of local labour markets. I measure labour market power with the wage-bill Herfindahl–Hirschman index (HHI),[1] which captures how payrolls are distributed among employers, and define local labour markets as 2-digit NAICS industries within provinces.

I begin with two country-level measures computed from administrative data for 2001–2019 (Figure 1). National concentration (NC) treats the country as a single labour market for each industry; aggregate local concentration (ALC) measures the concentration workers face in their industry–province market. Both declined over the period, but they diverged sharply during the Great Recession: NC rose moderately and briefly, while ALC rose by 14 percent between 2007 and 2009 before resuming its decline.

I then decompose these changes. Economy-wide, the long-run decline in ALC reflects falling concentration within industries, partially offset by a shift of the wage bill toward more concentrated industries. Public Administration plays a central role. Over 2001–2019 it contributed -11.7 percent of base-year ALC through within-industry deconcentration, $+9.3$ percent through its rising wage-bill share and -2.5 percent through the covariance term; its net contribution of -5.0 percent accounts for 54 percent of the within-province decline. This deconcentration coincides with the federal restructuring announced in Budget 2012, which targeted 19,234 positions, half of them in four large portfolios.[2] During the Great Recession, the private sector contracted while public employment held up, so the wage bill shifted toward this highly concentrated sector: reallocation across industries explains 68 percent of the within-province rise in ALC, and Public Administration alone accounts for 80 percent of that rise.

Excluding the public sector isolates competition among private employers. The long-run decline in private-sector ALC is robust (Table 7), but the two forces now reinforce each other: falling concentration within industries accounts for 88 percent of the within-province decline, and the shift of the wage bill toward less concentrated industries for 47 percent, partly offset by a covariance term of -36 percent. Information and cultural industries (61.5 percent), Utilities (41.0 percent) and Health care (26.3 percent) drive most of the decline. During the Great Recession, private-sector ALC rose by 13.9 percent between 2007 and 2010, and the within-industry component accounts for 75 percent of the within-province increase (Table 8). Health care (26.1 percent), Utilities (20.5 percent) and Retail trade (14.7 percent) lead this increase.

Firm turnover shapes both episodes. Among firms present in both 2001 and 2019, the within-province decline is about 35 percent larger than in the full sample: net entry raised concentration and offset roughly a quarter of the deconcentration generated by continuing firms. During the recession, turnover worked in the same direction but now amplified the rise: concentration among firms that survived 2007–2010 rose by 12.0 percent, against 13.9 percent in the full sample.

[1] The HHI is the standard concentration measure in antitrust enforcement. Recent policy debate emphasizes labour market concentration as a source of employer power; see Council of Economic Advisers (2016).

[2] Treasury Board President Tony Clement stated in November 2012: "We said we would eliminate 19,200 positions within the federal government and we are doing what we promised to do" (Treasury Board of Canada Secretariat, 2012).

I next ask whether concentration matters for pay. Using the employment-weighted mean of AKM firm effects as a market-level wage premium and a leave-one-out instrument for concentration (Azar et al., 2022; Rinz, 2022), I estimate an elasticity of the wage premium with respect to employment-based concentration of -0.028 in the specification with market and year fixed effects (Table 11). Finally, I embed an endogenous wage markdown in a two-agent New Keynesian (TANK) model calibrated to Canada. A government spending expansion tightens the labour market, narrows the markdown and shifts income from profits to wages, more strongly than in a model without cyclical monopsony.

Related literature. The paper contributes to four strands of literature. First, recent research documents diverging trends in national and local labour market concentration. Autor et al. (2020) document the rise of "superstar firms", which raised national concentration. Rinz (2022) shows, with employment-based HHIs, that local concentration in the United States declined from 1976 to 2015 while national concentration rose again after 1990. Handwerker and Dey (2021) document that the Great Recession raised concentration both locally and nationally in the United States, while the post-recession evidence is mixed (Yeh et al., 2022; Rossi-Hansberg et al., 2021; Rinz, 2022). Hershbein et al. (2020), using vacancy-based HHIs, and Manning and Petrongolo (2024) report declining local concentration in the United States and the United Kingdom. These studies generally associate higher local concentration with lower wages, but they do not decompose the sources of concentration change. I document the Canadian patterns using the wage-bill HHI, include all employers, including the public sector, and quantify the sources of change.

Second, the decomposition draws on the productivity literature. Foster et al. (2001) decompose aggregate productivity growth into within-firm, between-firm and entry–exit components; I adapt this framework to labour market concentration to separate within-industry competition, sectoral reallocation and firm turnover.

Third, a growing literature estimates the effect of employer concentration on wages. Azar et al. (2022) use a leave-one-out instrument and find that higher concentration lowers posted wages, and Rinz (2022) reports an earnings elasticity of about -0.03 with a similar design. I use as outcome the market-level wage premium from an AKM decomposition (Abowd et al., 1999; Card et al., 2013), which isolates the firm-side component of pay, and estimate the firm effects on a connected set that includes public employers.

Fourth, the quantitative section relates to models in which employer market power shapes the transmission of fiscal policy. I build on the TANK model of Cantore and Freund (2021) and on the cyclical monopsony mechanism of Bredemeier et al. (2026), and connect it to the concentration measures of Berger et al. (2022).

The remainder of the paper is organized as follows. Section 2 describes the data and measurement. Section 3 presents the concentration measures and the decomposition framework. Section 4 presents the results for the whole economy and for the private sector. Section 5 examines the role of entry, exit and continuing firms. Section 6 estimates the relationship between concentration and firm pay. Section 7 studies fiscal policy in a TANK model with employer market power, and Section 8 concludes.

2 Data and Measurement

Measuring labour market power. I measure labour market power with the wage-bill HHI, which captures how payroll is distributed among employers:

$$\text{HHI}_m = \sum_j s_{jm}^2, \qquad s_{jm} = \frac{w_{jm} n_{jm}}{\sum_{j'} w_{j'm} n_{j'm}}, \tag{1}$$

where w_{jm} denotes average earnings and n_{jm} employment of firm j in market m, so that s_{jm} is firm j's share of the market's wage bill. The wage-bill HHI reflects how earnings and workforce size jointly determine employer weight: a higher value means that a few firms account for a large share of wage payments in the market.

Business–Employee Analytical Microdata (BEAM). BEAM, developed by Statistics Canada, is a linked employer–employee dataset that connects worker and firm records through Social Insurance Numbers and Business Numbers. It contains annual information on earnings, employment, location, industry and employer. A key advantage for this analysis is that payroll records link each worker's earnings to the province of employment, which allows concentration to be measured where jobs are located.

Local labour markets. A labour market is defined by two features: workers are more attached to one market than to others, although some mobility is possible, and firms within a market compete to hire the same workers. Given the granularity of BEAM, I define a local labour market as a 2-digit NAICS industry within a province.

Sample. The goal is a comprehensive assessment of labour market power across the Canadian economy, so I apply minimal restrictions. The sample includes all workers with a non-missing employer identifier, strictly positive T4 earnings, and non-missing industry and province of employment, in all provinces and territories, from 2001 to 2019. Because concentration is measured on payroll rather than headcount, all employees are kept regardless of part-time or full-time status, as each contributes to the employer's wage bill.

Table 1 lists the 2-digit NAICS industries with their shares of the national wage bill over the whole sample and in 2001, 2007, 2009 and 2019. Table 2 reports the same shares by province and territory.

3 Empirical Framework

This section defines the two country-level concentration measures and the framework used to decompose their changes over time.

Table 1: Wage-Bill Shares by NAICS 2-Digit Sector

Code	Sector	Share of wage bill (%)				
		All	2001	2007	2009	2019
11	Agriculture, forestry, fishing and hunting	1.00	1.12	0.95	0.91	1.11
21	Mining, quarrying, and oil and gas extraction	2.96	2.36	3.34	2.82	2.67
22	Utilities	1.39	1.39	1.38	1.51	1.26
23	Construction	6.97	5.17	6.57	6.93	7.38
31–33	Manufacturing	13.53	19.03	15.05	12.76	11.01
41	Wholesale trade	5.85	5.84	6.15	5.77	5.80
44–45	Retail trade	7.00	6.96	6.97	7.12	6.89
48–49	Transportation and warehousing	4.52	4.65	4.46	4.27	4.78
51	Information and cultural industries	2.90	3.38	3.21	3.00	2.86
52	Finance and insurance	6.50	6.13	6.63	6.24	6.74
53	Real estate and rental and leasing	1.69	1.64	1.70	1.61	1.79
54	Professional, scientific and technical services	7.27	6.55	6.67	7.31	8.34
55	Management of companies and enterprises	0.94	1.06	1.03	1.01	0.75
56	Administrative and support, waste management	3.59	3.02	3.80	3.61	3.59
61	Educational services	7.51	7.31	7.23	7.64	7.51
62	Health care and social assistance	8.17	7.47	8.10	8.82	8.27
71	Arts, entertainment and recreation	0.96	1.00	0.95	1.00	0.97
72	Accommodation and food services	2.85	2.62	2.68	2.75	3.13
81	Other services (except public administration)	2.97	2.91	2.87	3.04	2.92
91	Public administration	11.41	10.40	10.25	11.85	12.22
Total		**100.00**	**100.00**	**100.00**	**100.00**	**100.00**

Notes: Wage-bill shares sum to 100 across sectors in each column. "All" covers 2001–2019. 2001 and 2019 are the first and last years of the sample; 2007 and 2009 are the pre-recession and recession years used in the economy-wide analysis.

Table 2: Wage-Bill Shares by Province and Territory

Code	Province/Territory	Share of wage bill (%)				
		All	2001	2007	2009	2019
10	Newfoundland and Labrador	1.2	1.11	1.1	1.2	1.23
11	Prince Edward Island	0.3	0.31	0.3	0.3	0.32
12	Nova Scotia	2.2	2.38	2.2	2.3	2.10
13	New Brunswick	1.8	1.86	1.8	1.9	1.66
24	Quebec	20.1	21.12	19.7	19.9	20.36
35	Ontario	40.5	42.85	40.7	39.7	40.54
46	Manitoba	3.1	3.07	3.0	3.2	3.10
47	Saskatchewan	2.8	2.44	2.6	2.9	2.75
48	Alberta	14.9	12.15	15.5	15.5	14.26
59	British Columbia	12.4	12.26	12.6	12.4	13.15
60	Yukon	0.1	0.12	0.1	0.1	0.15
61	Northwest Territories	0.2	0.23	0.3	0.2	0.21
62	Nunavut	0.1	0.10	0.1	0.1	0.17
	Total	**100.0**	**100.0**	**100.0**	**100.0**	**100.0**

Notes: Wage-bill shares sum to 100 in each column. "All" covers 2001–2019.

3.1 National and Aggregate Local Concentration

National concentration (NC). NC measures the average concentration across industries when each industry is treated as a single national labour market:

$$HHI^{NC} = \sum_{k \in \Omega} s_k \, HHI_k, \tag{2}$$

where HHI_k is the concentration of industry k computed over the whole country and s_k is the industry's share of the national wage bill.

Aggregate local concentration (ALC). Workers search for jobs mostly near where they live,[3] so the relevant concentration is that of the local market. ALC aggregates the concentration of each industry within each province:

$$HHI^{ALC} = \sum_{l} s_l \sum_{k} s_{kl} \, HHI_{kl}, \tag{3}$$

where HHI_{kl} is the concentration of industry k in province l, s_{kl} is the industry's share of the province's wage bill and s_l is the province's share of the national wage bill.

ALC systematically exceeds NC, because a local market contains only a subset of the firms operating in an industry nationally. In the United States, national HHIs range from 0.01 to 0.02, while local HHIs aggregated over commuting zones range from 0.11 to 0.13 (Rinz, 2022); in Canada, NC lies between 0.019 and 0.025 and ALC between 0.048 and 0.057 over 2001–2019 (Figure 1). ALC thus reflects the concentration workers face in their local industry, while NC abstracts from geography.

3.2 Decomposition Framework

I decompose the change in ALC into contributions from shifts across provinces and from changes within provinces, following the approach of Foster et al. (2001) for aggregate productivity growth.

Geographical decomposition. Equation (4) splits the change in ALC into a within-province effect, a between-province effect and a covariance effect:

$$\Delta HHI_t^{ALC} = \underbrace{\sum_{l} s_{l,t-1} \, \Delta HHI_{lt}}_{\text{Within-province}} + \underbrace{\sum_{l} HHI_{l,t-1} \, \Delta s_{lt}}_{\text{Between-province}} + \underbrace{\sum_{l} \Delta s_{lt} \, \Delta HHI_{lt}}_{\text{Covariance}}, \tag{4}$$

where $HHI_{lt} = \sum_k s_{klt} HHI_{klt}$ is the concentration of province l. The within-province effect captures changes in concentration inside each province. The between-province effect captures shifts of the wage bill across provinces: if provinces with high initial concentration gain wage-bill share, aggregate concentration rises. The covariance effect captures the co-movement between a province's size and its change in concentration.

[3] Marinescu and Rathelot (2018) show that job seekers are much less likely to apply to jobs farther from their residence.

Industry decomposition within provinces. Equation (5) further splits each province's change into a within-industry effect, a between-industry effect and a covariance effect:

$$\Delta HHI_{lt} = \underbrace{\sum_k s_{kl,t-1}\,\Delta HHI_{klt}}_{\text{Within-industry}} + \underbrace{\sum_k HHI_{kl,t-1}\,\Delta s_{klt}}_{\text{Between-industry}} + \underbrace{\sum_k \Delta s_{klt}\,\Delta HHI_{klt}}_{\text{Covariance}}. \tag{5}$$

The within-industry effect measures how concentration evolves inside each local industry. The between-industry effect captures how changes in the industrial composition of a province's wage bill move its concentration. The covariance effect captures whether industries whose concentration rises are also those whose wage-bill share grows.

Substituting Equation (5) into the within-province effect of Equation (4) gives a nested decomposition of the change in ALC.[4] Throughout, changes are expressed as fractions of the base-year ALC, so that a value of -0.075 is a 7.5 percent decline.

4 Results

I first document concentration for the whole economy, including the public sector, and then turn to the private sector. For each, I describe the trends in NC and ALC, use one-at-a-time counterfactuals to identify the main forces, and quantify them with the decomposition framework, over 2001–2019 and during the Great Recession.

4.1 Economy-Wide Labour Market Concentration

4.1.1 Trends

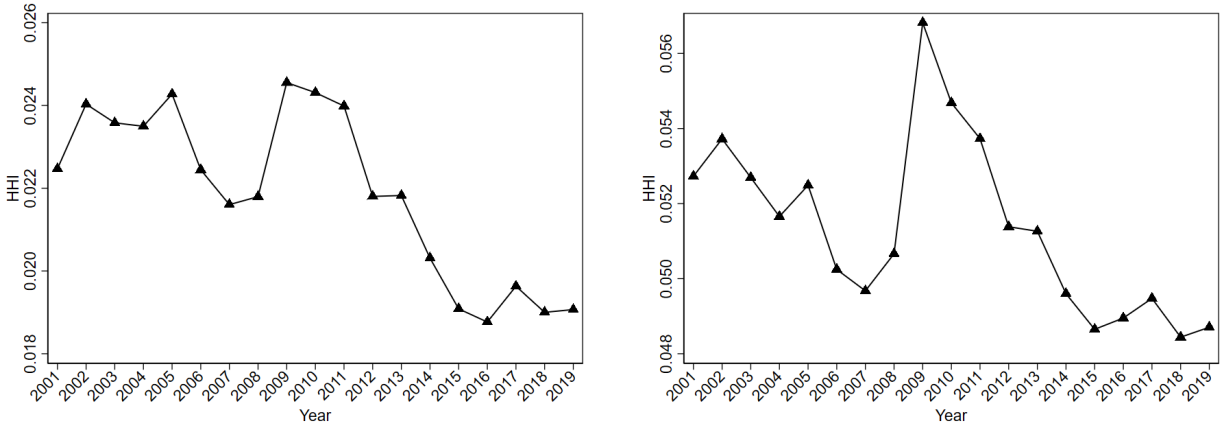
Figure 1 presents the trends. NC (panel a) was broadly stable before the recession, rose during the Great Recession and declined steadily from 2009, falling by 15.8 percent over the period. ALC (panel b) declined gradually from 2001 to 2019 but rose sharply during the recession, by 14 percent between 2007 and 2009, before resuming its decline and reaching its lowest level in 2019. Measures based on national markets thus display a different time pattern from those based on local markets.

4.1.2 Counterfactual Analysis

To understand why NC and ALC evolved differently, I construct counterfactual series (Figure 2) from the components of Equations (2) and (3). Each counterfactual holds all components at their 2001 values except one, which follows its observed path.

[4] Each term is summed over all provinces and industries, so the effects are aggregate: they measure the contribution to national ALC, not to a particular province or industry.

Figure 1: Economy-Wide Labour Market Concentration, 2001–2019



(a) National concentration (NC)

(b) Aggregate local concentration (ALC)

Notes: Panel (a) shows national concentration (NC), the wage-bill-weighted average of industry HHIs computed over the whole country (Equation 2). Panel (b) shows aggregate local concentration (ALC), the wage-bill-weighted average of industry HHIs computed within each province (Equation 3). All industries, including Public Administration.

Source: Author's calculations using BEAM microdata, 2001–2019.

Long run, 2001–2019. For NC (panel a), letting only industry concentration (HHI_k) vary produces a clear decline after 2009 that closely tracks actual NC. Letting only industrial composition (s_k) vary produces a mild upward trend from about 2012. Competition within industries was therefore the main force reducing national concentration, while the growing weight of concentrated industries partly offset it.

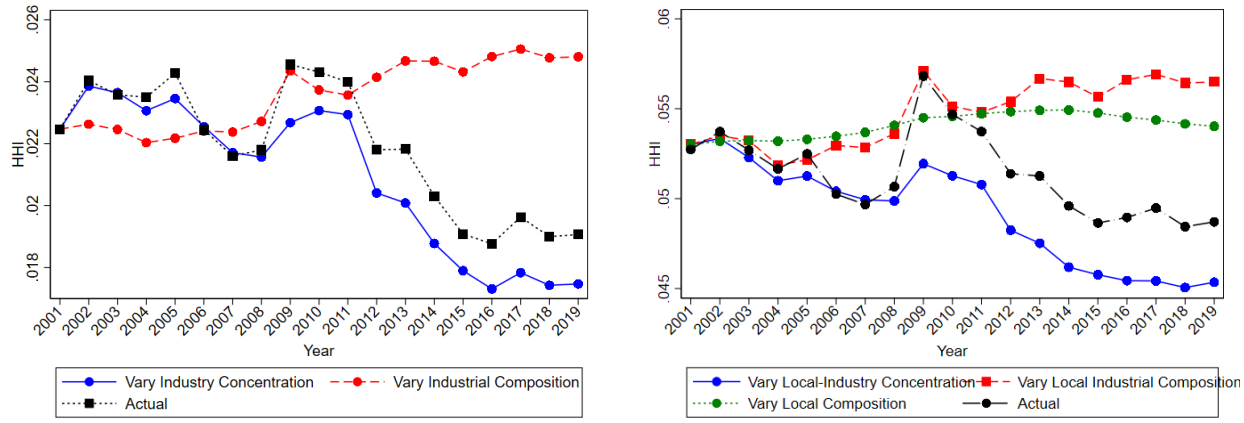
For ALC (panel b), letting only local-industry concentration (HHI_{kl}) vary produces a pronounced decline: firms became less dominant within local industries. Letting only local industrial composition (s_{kl}) vary produces a steady upward shift, as more concentrated industries gained weight. Letting only provincial composition (s_l) vary leaves ALC almost flat. ALC thus declined because concentration fell within local industries, partly offset by the growing weight of concentrated industries, while changes in the provincial distribution of payroll mattered little.

Great Recession, 2007–2009. For NC, letting only HHI_k vary leaves national concentration roughly flat in 2007–2008, whereas letting only s_k vary produces a sharp increase that closely matches the actual series. For ALC, the within-industry counterfactual is flat in 2007–2008 and rises only modestly in 2009, while the composition counterfactual closely matches the sharp increase in actual ALC. Both spikes therefore came mainly from sectoral reallocation: the recession shifted employment and wages toward industries that are structurally more concentrated. Table 3 summarizes these patterns; the decomposition below quantifies them.

4.2 Decomposing Economy-Wide Local Concentration

Long run, 2001–2019. Over 2001–2019, economy-wide ALC fell by 7.6 percent of its 2001 level (Table 4; detailed results in Appendix Table A2). The within-province effect (-0.092) is larger in magnitude than

Figure 2: Counterfactual Economy-Wide Concentration, 2001–2019



(a) National concentration (NC)

(b) Aggregate local concentration (ALC)

Notes: Each counterfactual holds all components of Equation 2 (panel a) or Equation 3 (panel b) at their 2001 values, except the one named in the legend, which follows its observed path. All industries, including Public Administration.

Source: Author's calculations using BEAM microdata, 2001–2019.

Table 3: Drivers of Economy-Wide Concentration: Counterfactual Evidence

Component	Long run (2001–2019)			Great Recession (2007–2009)		
	National	Local		National	Local	
	Effect	Effect	Magnitude	Effect	Effect	Magnitude
Within-industry concentration	↓	↓	Strong	→	↑	Weak
Industry composition (within provinces)	↑	↑	Moderate	↑	↑	Strong
Geographic distribution of payroll	–	→	Negligible	–	↑	Moderate
Aggregate effect	↓	↓	Moderate	↑	↑	Strong

Notes: Summary of the counterfactual series in Figure 2. Arrows give the direction of each component's contribution: ↓ lowers concentration, ↑ raises it and → indicates little or no effect. "Industry composition" refers to changes in the weight of each industry within provinces; "geographic distribution" to changes in each province's share of national payroll, which does not enter NC.

the total change, so shifts across provinces slightly raised concentration. Within provinces, falling concentration within industries was by far the largest force (-0.146): most industries became less concentrated internally. The between-industry effect partly offset this decline ($+0.067$), as the wage bill tilted toward more concentrated industries, and the covariance term was small (-0.013).

Great Recession, 2007–2009. The recession interrupted the gradual decline with a sharp but temporary rise. Between 2007 and 2009, ALC rose by 14.4 percent of its 2007 level, and 88.9 percent of this increase occurred within provinces (Table 5; Appendix Table A3). The between-industry component was the largest contributor ($+0.087$, or 68 percent of the within-province effect): activity shifted toward more concentrated industries. The within-industry component added $+0.040$ (32 percent), reflecting rising dominance of large employers within local industries, and the covariance term was negligible. Both compositional change and

Table 4: Economy-Wide ALC: Decomposition of the Long-Run Change (2001–2019)

Component	Effect	Value	Interpretation
Within-industry	↓	−0.146	Concentration fell within industries: wage bills became more evenly spread across employers.
Between-industry	↑	+0.067	The wage bill shifted toward industries with fewer, larger employers.
Covariance	↓	−0.013	Industries whose concentration fell tended to gain wage-bill share, adding slightly to the decline.
Within-province effect	↓	−0.092	Net decline within provinces, driven by within-industry deconcentration.
Total change in ALC	↓	−0.076	

Notes: Contributions to the change in economy-wide ALC, expressed as fractions of the 2001 ALC (Equations 4 and 5). The within-province effect is the sum of the first three rows; the difference between the total change and the within-province effect is the between-province effect. Industry detail in Appendix Table A2.

changes in firm dominance therefore contributed, with composition dominant.

Table 5: Economy-Wide ALC: Decomposition of the Change During the Great Recession (2007–2009)

Component	Effect	Value	Interpretation
Within-industry	↑	+0.040	Large employers gained weight within industries, through exit of smaller competitors, consolidation or faster growth of incumbents.
Between-industry	↑	+0.087	The wage bill shifted toward structurally concentrated sectors, above all Public Administration.
Covariance	→	+0.001	Little co-movement between sectoral growth and changes in concentration.
Within-province effect	↑	+0.128	Most of the rise occurred within provinces, through both reallocation and within-industry consolidation.
Total change in ALC	↑	+0.144	

Notes: Contributions to the change in economy-wide ALC, expressed as fractions of the 2007 ALC (Equations 4 and 5). The within-province effect is the sum of the first three rows. Industry detail in Appendix Table A3.

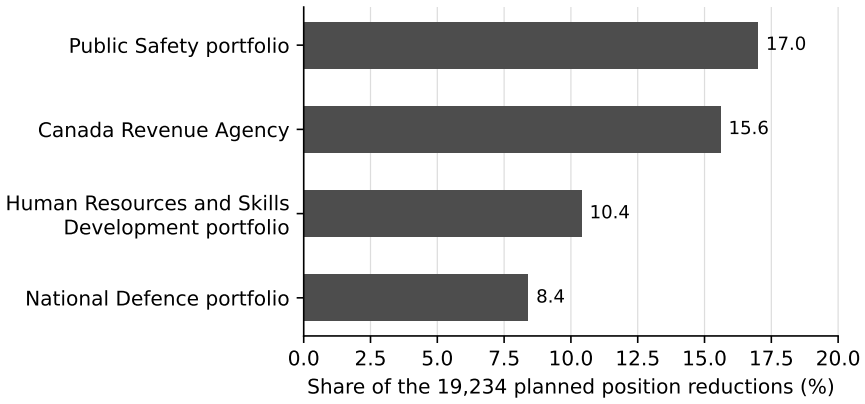
4.3 The Public Sector as a Driver of Economy-Wide Concentration

Breaking down the long-run within-province decline by industry (Appendix Table A2, Panel B) shows that Public Administration (NAICS 91) made by far the largest contribution. Its within-industry concentration fell by 11.7 percent of base-year ALC, its rising wage-bill share added 9.3 percent and the covariance term subtracted 2.5 percent; the net contribution of -0.050 accounts for 54 percent of the within-province decline. Information and cultural industries (23 percent), Utilities (17 percent) and Health care (10 percent) follow, while Finance and insurance (-9 percent) and Educational services (-7 percent) moved against the trend. Among firms present in both 2001 and 2019, within-industry deconcentration in Public Administration is of similar size (-0.125, against -0.117 in the full sample; Appendix Table A4), so the decline came from continuing employers rather than from entry.

This pattern is consistent with the federal restructuring announced in Budget 2012. The government planned to eliminate 19,234 positions over three years, about 12,000 through workforce adjustment and 7,200 through attrition and reduced hiring (Treasury Board of Canada Secretariat, 2012). Four portfolios accounted for 51 percent of these planned reductions (Figure 3): Public Safety (17.0 percent), the Canada Revenue Agency (15.6 percent), Human Resources and Skills Development (10.4 percent) and National Defence (8.4 percent). Cuts concentrated among the largest public employers mechanically reduce concentration within the sector.

The national capital region offers a geographic signature of this restructuring. Public Administration employment among residents of the Ottawa–Gatineau census metropolitan area rose steadily from 2007 to 2012 and fell sharply in 2013 (Figure 4), as the Budget 2012 reductions took effect.

Figure 3: Planned Position Reductions in the Federal Public Service After Budget 2012: Four Largest Portfolios



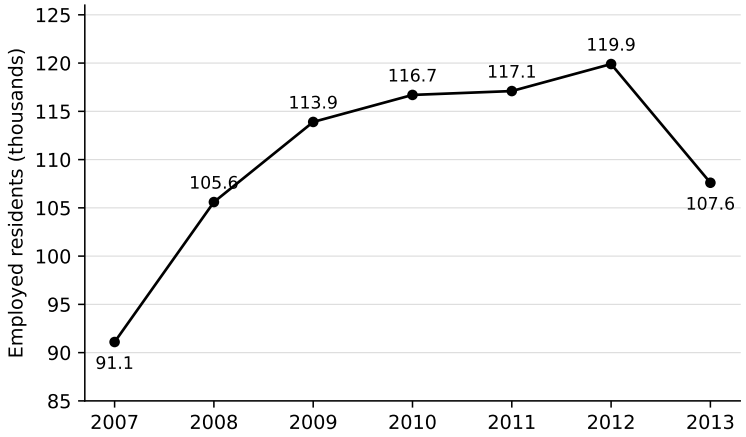
Notes: Share of the 19,234 full-time-equivalent positions that the federal government planned to eliminate over three years following Budget 2012, by portfolio. The four portfolios shown account for 51 percent of planned reductions.

Source: Treasury Board of Canada Secretariat (2012); author's calculations.

During the Great Recession, Public Administration played the opposite role. It accounts for 80 percent of the within-province increase in ALC between 2007 and 2009 (Appendix Table A3, Panel B). This reflects mainly a large between-industry effect (+0.093): as private employment contracted and public employment held up, a growing share of the wage bill moved into this highly concentrated sector. Within-industry concentration in the sector also rose slightly (+0.009). Public Administration also had by far the lowest firm turnover in the economy during the recession, with entry and exit rates of 6.4 and 5.7 percent against roughly 14 to 26 percent in private industries (Appendix Table A6).

Because the public sector dominates economy-wide dynamics, the rest of the analysis excludes it, in order to isolate concentration dynamics driven by competition among private employers.

Figure 4: Public Administration Employment in the Ottawa–Gatineau CMA, 2007–2013



Notes: Employed residents of the Ottawa–Gatineau census metropolitan area working in Public Administration (NAICS 91), in thousands. The series counts workers by place of residence, whereas BEAM records employment by province of work.

Source: City of Ottawa (2014), Table 12.

4.4 Private-Sector Labour Market Concentration

4.4.1 Trends

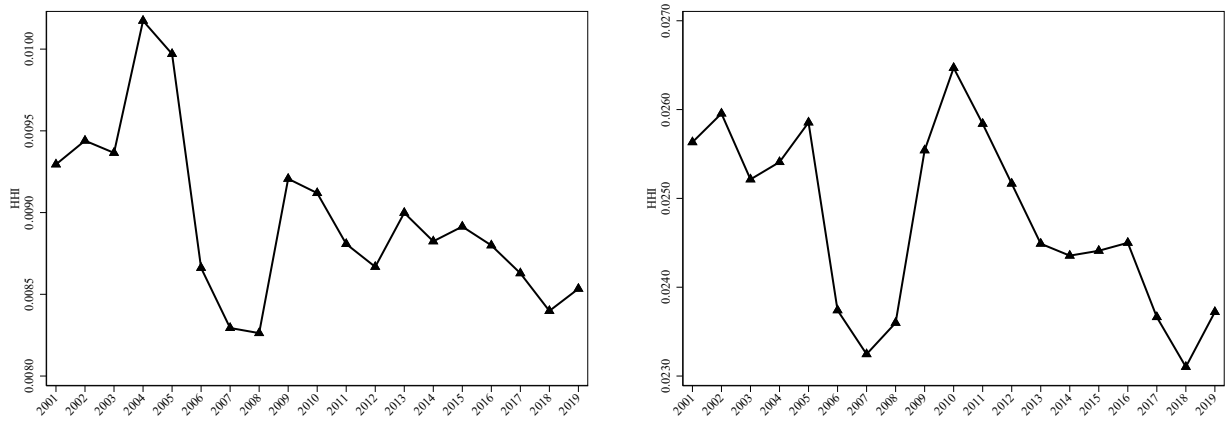
Figure 5 presents the private-sector trends. Without the public sector, NC follows a flatter path, declining by 8.3 percent over the period against 15.8 percent when the public sector is included, and its recession spike is smaller (10.9 percent against 12.2 percent). ALC is more robust to the exclusion: the long-run decline is about 7.5 percent in both cases, and the recession spike remains close to 14 percent. The main difference lies in timing: economy-wide ALC peaks in 2009 and declines immediately, whereas private-sector ALC keeps rising until 2010. The stability of public employment thus shortened the local concentration episode. For this reason, the private-sector recession window is 2007–2010.

4.4.2 Counterfactual Analysis

Figure 6 repeats the counterfactual exercise for the private sector.

Long run, 2001–2019. For NC (panel a), the within-industry counterfactual closely tracks the actual decline, particularly after 2009, while varying only industrial composition produces a nearly flat path from 2006 onward. Greater competition within industries was thus the main force reducing national concentration. For ALC (panel b), letting only local-industry concentration vary produces a pronounced decline through 2018, confirming that within-industry competition was the primary driver. Varying local industrial composition yields a gradual decline after 2010, a modest additional contribution from the shift toward less concentrated industries. Varying only provincial composition leaves ALC nearly flat.

Figure 5: Private-Sector Labour Market Concentration, 2001–2019



(a) National concentration (NC)

(b) Aggregate local concentration (ALC)

Notes: Panel (a) shows NC and panel (b) ALC, computed as in Figure 1 but excluding Public Administration (NAICS 91).
Source: Author's calculations using BEAM microdata, 2001–2019.

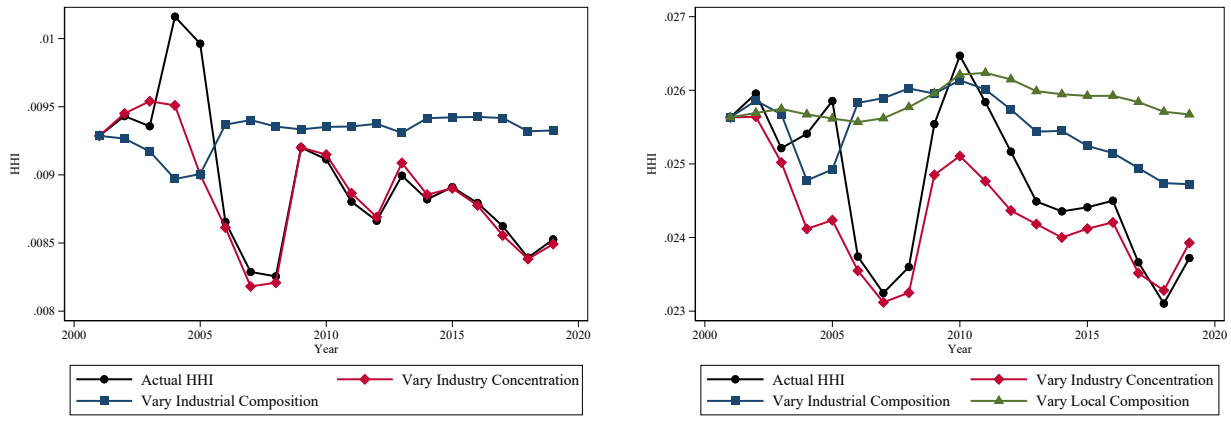
Great Recession, 2007–2010. For NC, the recession-era increase was driven by rising within-industry concentration: when only HHI_k varies, national concentration rises sharply over 2007–2009 and tracks the actual series, while composition alone produces no meaningful change. For ALC, letting only local-industry concentration vary reproduces most of the observed spike, local industrial composition plays a minor role and the provincial distribution of payroll contributes modestly. The within-industry counterfactual does not fully reach the actual peak, because the one-at-a-time approach omits the interaction between rising concentration and simultaneous shifts in composition. Table 6 summarizes these patterns, and the decomposition below quantifies each term, including the covariance.

Table 6: Drivers of Private-Sector Concentration: Counterfactual Evidence

Component	Long run (2001–2019)			Great Recession (2007–2010)		
	National	Local		National	Local	
	Effect	Effect	Magnitude	Effect	Effect	Magnitude
Within-industry concentration	↓	↓	Strong	↑	↑	Strong
Industry composition (within provinces)	→	↓	Moderate	→	↑	Weak
Geographic distribution of payroll	–	→	Negligible	–	↑	Moderate
Aggregate effect	↓	↓	Moderate	↑	↑	Strong

Notes: Summary of the counterfactual series in Figure 6. Arrows give the direction of each component's contribution: ↓ lowers concentration, ↑ raises it and → indicates little or no effect. Public Administration (NAICS 91) is excluded.

Figure 6: Counterfactual Private-Sector Concentration, 2001–2019



(a) National concentration (NC)

(b) Aggregate local concentration (ALC)

Notes: Each counterfactual holds all components at their 2001 values except the one named in the legend. Public Administration (NAICS 91) is excluded.

Source: Author's calculations using BEAM microdata, 2001–2019.

4.5 Decomposing Private-Sector Local Concentration

4.5.1 Long Run, 2001–2019

Over 2001–2019, private-sector ALC declined by 7.5 percent of its 2001 level (Table 7, Panel A). The within-province effect accounts for virtually all of this decline (-0.075), so changes in the provincial distribution of payroll played no role. Within provinces, two forces reinforced each other. The within-industry component contributed -0.067 (88.4 percent of the within-province effect): most industries became less concentrated internally. The between-industry component contributed -0.035 (47.1 percent): the wage bill shifted toward less concentrated industries. A positive covariance term ($+0.027$) partly offset these forces: industries whose concentration fell tended to lose wage-bill share.

Industry contributions. Panel B of Tables 7 and 9 reports industry contributions for all firms and for firms present in both 2001 and 2019. The decline was concentrated in a few sectors, with Information and cultural industries alone accounting for 61.5 percent of the within-province effect, and three industries illustrate distinct mechanisms.

Health care accounted for 26.3 percent of the decline (-0.020), driven by a large within-industry effect (-0.027). Among continuing firms, however, its contribution changes sign ($+0.003$): deconcentration in health care came from entry. The sector had one of the highest net entry rates over the period (35 percent), consistent with provincial policies that expanded private capacity in health services.[5] New employers gave health-care workers, such as nurses, more options and spread the sector's wage bill across more firms.

[5]Several provinces allow physicians to practise in both the public and the private systems, and amendments to provincial legislation have expanded private-sector capacity; see `https://pmc.ncbi.nlm.nih.gov/articles/PMC3091914/`.

Manufacturing contributed 13.3 percent (-0.010) through a different channel: its within-industry concentration rose ($+0.011$), but the decline of its wage-bill share produced a larger negative between-industry effect (-0.022). Among continuing firms, the within-industry effect is negative (-0.010). The rise in full-sample concentration therefore reflects the exit of smaller manufacturers (a net entry rate of -16 percent), which raised the weight of the remaining large employers.

Some sectors moved against the overall trend. Finance and insurance was the largest concentrating force ($+0.020$, offsetting 26.9 percent of the within-province decline), driven by its growing weight in the economy (a between-industry effect of $+0.017$). Among continuing firms, its within-industry effect is strongly negative (-0.022): incumbents became less dominant, and the concentrating contribution came from compositional change rather than from consolidation among existing firms. Educational services also raised concentration ($+0.019$), mostly within the industry.

Table 7: Decomposition of Private-Sector ALC Change, Long Run (2001–2019)

Panel A: Overall decomposition

Total change	Within-province	Within-industry	Between-industry	Covariance
-0.075	-0.075 (100.9%)	-0.067 (88.4%)	-0.035 (47.1%)	0.027 (-35.6%)

Panel B: Industry contributions to the within-province effect

Industry (NAICS)	Within-industry	Between-industry	Covariance	Total	Share (%)
Agriculture, forestry & fishing (11)	0.002	0.000	0.001	0.003	-3.7
Mining & oil and gas extraction (21)	-0.007	0.013	-0.005	0.001	-1.3
Utilities (22)	-0.017	-0.015	0.001	-0.031	41.0
Construction (23)	-0.005	0.003	0.000	-0.002	3.2
Manufacturing (31–33)	0.011	-0.022	0.001	-0.010	13.3
Wholesale trade (41)	0.003	0.000	0.000	0.003	-4.5
Retail trade (44–45)	-0.003	0.000	0.000	-0.003	4.0
Transportation & warehousing (48–49)	-0.020	0.002	0.000	-0.017	23.0
Information & cultural industries (51)	-0.028	-0.028	0.010	-0.046	61.5
Finance & insurance (52)	0.003	0.017	0.001	0.020	-26.9
Real estate (53)	0.000	0.000	0.000	0.001	-0.8
Professional services (54)	0.000	0.003	0.000	0.003	-4.5
Management of companies (55)	0.008	-0.004	-0.003	0.002	-2.9
Administrative services (56)	-0.001	0.001	0.000	0.000	-0.2
Educational services (61)	0.014	0.006	-0.001	0.019	-25.1
Health care (62)	-0.027	-0.014	0.021	-0.020	26.3
Arts & entertainment (71)	-0.002	0.000	0.000	-0.001	1.8
Accommodation & food (72)	0.000	0.001	0.000	0.001	-0.7
Other services (81)	0.002	0.000	0.000	0.003	-3.5
Within-province total				**-0.075**	**100.0**

Notes: All values are changes in aggregate local concentration (ALC) expressed as fractions of the 2001 ALC; for example, -0.075 is a 7.5 percent decline. Panel A splits the total change into the within-province effect and, within it, the within-industry, between-industry and covariance terms (Equations 4 and 5); the remainder of the total change is the between-province effect. Percentages in parentheses are shares of the immediate parent term. Panel B allocates the within-province effect across industries; the last numeric column is each industry's share of the within-province effect. Values are rounded to three decimals. Public Administration (NAICS 91) is excluded from the sample.

4.5.2 Great Recession, 2007–2010

Between 2007 and 2010, private-sector ALC rose by 13.9 percent of its 2007 level (Table 8, Panel A). The within-province effect accounts for 77.3 percent of this increase (+0.107), and between-province shifts for the remainder. Within provinces, the increase came mainly from the within-industry component (+0.081, 75.2 percent of the within-province effect): concentration rose inside local industries rather than through compositional shifts. The between-industry effect was small (+0.010) and the covariance term contributed +0.016.

Industry contributions. The increase was broad-based (Table 8, Panel B). Health care was the largest contributor (26.1 percent, +0.028), with all three margins positive: within-industry (+0.010), between-industry (+0.009) and covariance (+0.009). The sector both gained wage-bill share and became more concentrated within local markets. Among continuing firms, its contribution falls to 10.4 percent (Table 10): the recession-era rise reflected sectoral expansion and turnover rather than consolidation among incumbents. Utilities (20.5 percent) and Retail trade (14.7 percent) follow. Finance and insurance contributed 10.2 percent (+0.011), almost entirely within the industry; among continuing firms, its contribution falls to 3.1 percent, so turnover amplified the increase in this sector. Manufacturing contributed 8.8 percent (+0.009): a large within-industry effect (+0.017) was partly offset by the decline of its wage-bill share (−0.007). Among continuing firms, the within-industry effect remains large (+0.015), which points to consolidation among incumbent manufacturers during the recession, the opposite of their long-run pattern.

5 Entry, Exit and Continuing Firms

This section examines the firm-level sources of concentration change, entry, exit and incumbent growth, and quantifies how turnover shaped ALC.

5.1 Sources of Change in Concentration

Changes in the wage-bill HHI result from three mechanisms:[6] firm exit, firm entry and the growth of incumbent firms.

Exit. Exit raises concentration when smaller firms leave, because their wage-bill share is redistributed toward the remaining, larger employers. The exit rate in market m between $t-1$ and t is

$$\text{ExitRate}_{mt} = \frac{X_{mt}}{F_{m,t-1}},$$

where X_{mt} is the number of firms exiting between $t-1$ and t and $F_{m,t-1}$ the number of firms at the start of the period.

[6] This follows the tradition of reallocation studies such as Baily et al. (1992) and Foster et al. (2001).

Table 8: Decomposition of Private-Sector ALC Change, Great Recession (2007–2010)

Panel A: Overall decomposition

Total change	Within-province	Within-industry	Between-industry	Covariance
0.139	0.107 (77.3%)	0.081 (75.2%)	0.010 (9.5%)	0.016 (15.4%)

Panel B: Industry contributions to the within-province effect

Industry (NAICS)	Within-industry	Between-industry	Covariance	Total	Share (%)
Agriculture, forestry & fishing (11)	0.000	0.000	0.000	0.000	0.3
Mining & oil and gas extraction (21)	−0.009	−0.003	0.001	−0.011	−9.8
Utilities (22)	0.006	0.018	−0.001	0.022	20.5
Construction (23)	0.003	0.001	0.000	0.004	3.4
Manufacturing (31–33)	0.017	−0.007	0.000	0.009	8.8
Wholesale trade (41)	0.000	−0.001	0.000	−0.001	−1.1
Retail trade (44–45)	0.006	0.001	0.008	0.016	14.7
Transportation & warehousing (48–49)	0.013	−0.003	0.000	0.009	8.8
Information & cultural industries (51)	0.017	−0.012	−0.001	0.004	3.3
Finance & insurance (52)	0.012	−0.001	0.000	0.011	10.2
Real estate (53)	0.000	0.000	0.000	0.000	0.2
Professional services (54)	0.005	0.001	0.001	0.007	6.2
Management of companies (55)	0.002	−0.001	0.000	0.002	1.5
Administrative services (56)	−0.004	−0.001	0.001	−0.004	−3.8
Educational services (61)	0.001	0.008	0.000	0.009	8.7
Health care (62)	0.010	0.009	0.009	0.028	26.1
Arts & entertainment (71)	0.000	0.001	0.000	0.000	0.2
Accommodation & food (72)	−0.001	0.000	0.000	−0.001	−0.9
Other services (81)	0.002	0.000	0.000	0.003	2.7
Within-province total				**0.107**	**100.0**

Notes: All values are changes in aggregate local concentration (ALC) expressed as fractions of the 2007 ALC; for example, −0.075 is a 7.5 percent decline. Panel A splits the total change into the within-province effect and, within it, the within-industry, between-industry and covariance terms (Equations 4 and 5); the remainder of the total change is the between-province effect. Percentages in parentheses are shares of the immediate parent term. Panel B allocates the within-province effect across industries; the last numeric column is each industry's share of the within-province effect. Values are rounded to three decimals. Public Administration (NAICS 91) is excluded from the sample.

Entry. Entry generally lowers concentration by adding competitors that dilute the wage-bill shares of existing firms; its effect depends on the size of entrants. The entry rate is

$$\text{EntryRate}_{mt} = \frac{N_{mt}}{F_{mt}},$$

where N_{mt} is the number of firms entering during period t and F_{mt} the number of firms at the end of the period.

Incumbent growth. Growth among incumbents changes the distribution of wage-bill shares: if large incumbents grow faster, concentration rises; if smaller incumbents catch up, it falls. The contribution of continuing firms is

$$\Delta HHI_{mt}^{\text{inc}} = \sum_{i \in C_m} \left(s_{imt}^2 - s_{im,t-1}^2 \right),$$

where C_m is the set of firms present in market m in both periods.

These firm-level changes act first within local industries, but they propagate to ALC through Equation (3): they change the concentration of each local industry, which can alter the industrial composition of provinces

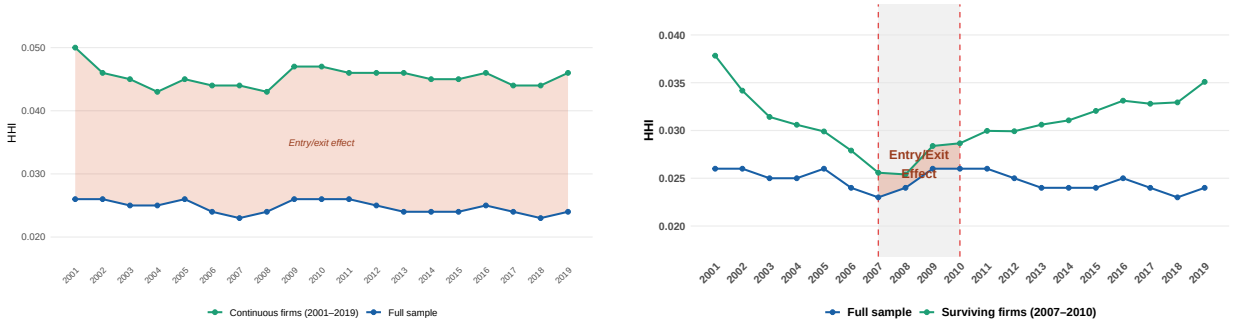
and, ultimately, the distribution of activity across provinces. Comparing the decompositions for all firms and for continuing firms quantifies the net contribution of entry and exit.

5.2 The Role of Entry, Exit and Continuing Firms

Long run, 2001–2019. Over the full period, private-sector ALC declined by 7.5 percent. Restricting the sample to firms present in both 2001 and 2019 yields a within-province decline of −0.102, about 35 percent larger than in the full sample (−0.075; Table 9). Net entry therefore offset the deconcentration generated by continuing firms: on balance, entering and exiting firms pushed concentration up, and they offset about 26 percent of the incumbents' decline. Panel (a) of Figure 7 shows that concentration among continuing firms is persistently higher than in the full sample, while both series decline over the period.

Great Recession, 2007–2010. During the recession, turnover again raised concentration, but this time it added to the rise. Among firms that survived the whole 2007–2010 window, ALC rose by 12.0 percent and the within-province effect was +0.100, compared with 13.9 percent and +0.107 in the full sample (Table 10). Entry and exit thus account for about one seventh of the recession-era increase. Panel (b) of Figure 7 shows that both series rise by similar proportions during 2007–2010 (about 12 percent among survivors and 13 percent in the full sample). The composition of the survivors' increase also differs: the within-province share is higher (83.1 percent against 77.3 percent), and within provinces the between-industry component is much larger (44.4 percent of the within-province effect, against 9.5 percent in the full sample). The economy-wide decomposition gives the same result: among firms that survived 2007–2009, ALC rose by 11.8 percent, against 14.4 percent for all firms (Appendix Table A5).

Figure 7: Concentration Among All Firms and Among Continuing Firms



(a) Long run, 2001–2019

(b) Great Recession, 2007–2010

Notes: Each panel plots private-sector ALC for the full sample and for a restricted set of firms: firms present in every year from 2001 to 2019 in panel (a), and firms present in every year from 2007 to 2010 in panel (b). The shaded area between the curves is the difference in measured concentration attributable to entering and exiting firms.

Source: Author's calculations using BEAM microdata, 2001–2019.

Table 9: Decomposition of Private-Sector ALC Change, Long Run (2001–2019): Continuing Firms

Panel A: Overall decomposition

Total change	Within-province	Within-industry	Between-industry	Covariance
−0.081	−0.102 (125.5%)	−0.108 (106.4%)	−0.039 (38.7%)	0.046 (−45.1%)

Panel B: Industry contributions to the within-province effect

Industry (NAICS)	Within-industry	Between-industry	Covariance	Total	Share (%)	Net entry (%)
Agriculture, forestry & fishing (11)	0.000	0.000	0.001	0.001	−0.6	−13.1
Mining & oil and gas extraction (21)	0.002	0.016	0.001	0.019	−18.8	18.4
Utilities (22)	−0.006	−0.018	0.000	−0.024	23.5	12.7
Construction (23)	−0.011	−0.002	0.003	−0.010	9.8	35.2
Manufacturing (31–33)	−0.010	−0.026	0.015	−0.020	20.1	−16.1
Wholesale trade (41)	0.005	0.001	0.001	0.006	−6.1	−10.7
Retail trade (44–45)	−0.010	−0.004	0.001	−0.013	12.5	−1.8
Transportation & warehousing (48–49)	−0.015	−0.001	0.000	−0.016	15.3	49.4
Information & cultural industries (51)	−0.048	−0.042	0.019	−0.071	70.0	16.1
Finance & insurance (52)	−0.022	0.035	−0.007	0.006	−5.5	15.8
Real estate (53)	0.001	0.000	0.001	0.002	−2.3	30.9
Professional services (54)	0.003	0.004	0.002	0.009	−8.8	35.5
Management of companies (55)	0.007	−0.004	0.001	0.005	−4.6	−152.4
Administrative services (56)	−0.002	0.001	0.000	−0.001	0.8	20.1
Educational services (61)	0.002	−0.003	−0.001	−0.002	1.9	30.6
Health care (62)	−0.009	0.006	0.005	0.003	−3.0	35.0
Arts & entertainment (71)	−0.002	−0.002	0.003	0.000	−0.1	12.0
Accommodation & food (72)	−0.001	0.000	0.000	−0.001	0.9	13.4
Other services (81)	0.005	0.000	0.000	0.005	−5.1	−6.3
Within-province total				**−0.102**	**100.0**	

Notes: All values are changes in aggregate local concentration (ALC) expressed as fractions of the 2001 ALC; for example, −0.075 is a 7.5 percent decline. Panel A splits the total change into the within-province effect and, within it, the within-industry, between-industry and covariance terms (Equations 4 and 5); the remainder of the total change is the between-province effect. Percentages in parentheses are shares of the immediate parent term. Panel B allocates the within-province effect across industries; the last numeric column is each industry's share of the within-province effect. Values are rounded to three decimals. Public Administration (NAICS 91) is excluded from the sample. The sample is restricted to firms present in both 2001 and 2019. Net entry is the number of entering firms minus the number of exiting firms over the window, as a percentage of the industry's firm count; it is reported for interpretation and is not part of the decomposition identity. Because the firm count used as denominator is not the count at the start of the window, the rate can fall below −100 percent when an industry's number of firms contracts sharply, as for Management of companies (NAICS 55).

6 Labour Market Concentration and Firm Pay Policy

This section examines whether higher employer concentration translates into lower firm-side pay. The empirical design addresses three sources of bias: worker composition, the omission of public employers and the joint determination of concentration and pay.

6.1 Empirical Strategy

Worker composition. Market-level earnings mix firm pay policies with the composition of the workforce: if workers sort across markets on productivity, a regression of earnings on concentration captures both. I therefore use as outcome the market-level wage premium $\overline{\psi}_{m,t}$, the employment-weighted mean of the AKM firm effects of the private-sector firms in market m and year t.

Table 10: Decomposition of Private-Sector ALC Change, Great Recession (2007–2010): Continuing Firms

Panel A: Overall decomposition

Total change	Within-province	Within-industry	Between-industry	Covariance
0.120	0.100 (83.1%)	0.051 (51.1%)	0.044 (44.4%)	0.005 (4.5%)

Panel B: Industry contributions to the within-province effect

Industry (NAICS)	Within-industry	Between-industry	Covariance	Total	Share (%)	Net entry (%)
Agriculture, forestry & fishing (11)	0.000	0.000	0.000	0.000	0.4	−1.5
Mining & oil and gas extraction (21)	−0.001	0.003	0.001	0.003	3.0	0.7
Utilities (22)	−0.017	0.037	−0.010	0.010	9.6	−3.5
Construction (23)	0.003	0.001	0.000	0.004	3.7	5.7
Manufacturing (31–33)	0.015	−0.008	0.000	0.008	8.1	−5.2
Wholesale trade (41)	0.000	−0.001	0.000	−0.001	−1.1	−4.9
Retail trade (44–45)	0.000	0.000	0.000	0.000	−0.1	−2.9
Transportation & warehousing (48–49)	0.012	−0.002	0.000	0.009	9.3	6.0
Information & cultural industries (51)	0.024	0.004	0.011	0.038	38.4	2.8
Finance & insurance (52)	0.007	−0.003	0.000	0.003	3.1	−1.6
Real estate (53)	0.001	0.000	0.000	0.000	0.3	5.7
Professional services (54)	0.005	0.001	0.001	0.007	7.4	6.3
Management of companies (55)	0.002	0.000	0.000	0.001	1.1	−5.3
Administrative services (56)	−0.005	−0.001	0.001	−0.004	−4.2	2.5
Educational services (61)	0.002	0.005	0.000	0.007	7.5	2.8
Health care (62)	0.002	0.008	0.000	0.010	10.4	8.1
Arts & entertainment (71)	0.000	0.000	0.000	0.001	0.8	−0.5
Accommodation & food (72)	−0.001	0.000	0.000	−0.001	−0.9	0.8
Other services (81)	0.003	0.000	0.001	0.003	3.3	−3.5
Within-province total				**0.100**	**100.0**	

Notes: All values are changes in aggregate local concentration (ALC) expressed as fractions of the 2007 ALC; for example, −0.075 is a 7.5 percent decline. Panel A splits the total change into the within-province effect and, within it, the within-industry, between-industry and covariance terms (Equations 4 and 5); the remainder of the total change is the between-province effect. Percentages in parentheses are shares of the immediate parent term. Panel B allocates the within-province effect across industries; the last numeric column is each industry's share of the within-province effect. Values are rounded to three decimals. Public Administration (NAICS 91) is excluded from the sample. The sample is restricted to firms present in every year from 2007 to 2010. Net entry is the number of entering firms minus the number of exiting firms over the window, as a percentage of the industry's firm count; it is reported for interpretation and is not part of the decomposition identity.

AKM specification. Firm wage premia are recovered from the two-way fixed effects model of Abowd et al. (1999). For worker i employed at firm $j(i,t)$ in year t,

$$y_{it} = \alpha_i + \psi_{j(i,t)} + X_{it}'\beta + \varepsilon_{it}, \tag{6}$$

where y_{it} is log real annual earnings, α_i captures time-invariant worker heterogeneity, $\psi_{j(i,t)}$ is the firm-specific wage premium, X_{it} contains a cubic polynomial in age and year-by-province fixed effects, and ε_{it} is an idiosyncratic residual. Because y_{it} is annual earnings, the firm effects capture both hourly pay and hours policies. Identification requires restricting the sample to the largest connected set of workers and firms linked through job-to-job mobility. Following Card et al. (2013) and Dostie et al. (2023), I first residualize y_{it} on X_{it} and then estimate the two-way fixed effects model on the residualized outcome with the iterative algorithm of Gaure (2013). Since ψ_j is measured in log points, $\overline{\psi}_{m,t}$ is the log wage premium of market m.

Public employers. AKM decompositions are usually estimated on private-sector samples. Public employers, however, are an important outside option for workers in several private industries, and excluding them from

the connected set can distort the estimated private-sector firm effects (Djolaud, 2026). I therefore estimate the AKM model on a connected set that pools private and public employment, so that firm effects are measured against workers' full set of employment opportunities. Public employers enter only at this stage: the wage premium $\overline{\psi}_{m,t}$ and the HHI are both computed over private-sector firms, the employers exposed to concentration-driven wage setting.

Endogenous concentration. Concentration and pay policies respond to the same local shocks. A positive demand or productivity shock can lead incumbents to raise pay premia and attract entrants that lower concentration, while adverse shocks can compress premia and trigger exit. Other mechanisms, such as measurement error in market-level HHIs, work in the opposite direction, so the sign of the OLS bias is ambiguous. Following Azar et al. (2022) and Rinz (2022), I instrument the log HHI of market m (industry k in province l) with a leave-one-out measure of concentration in the same industry in all other provinces:

$$HHI_{k,t}^{-l} = \frac{\sum_{z \neq l} HHI_{zk,t}\, Emp_{zk,t}}{\sum_{z \neq l} Emp_{zk,t}}, \tag{7}$$

where z indexes provinces and $Emp_{zk,t}$ is industry–province employment. The instrument captures variation in concentration driven by industry-wide forces, such as technological change, regulation or changes in industrial organization, that are common across provinces. The exclusion restriction requires that these forces affect a market's wage premium only through local concentration, conditional on the fixed effects; industry-wide shocks that move pay premia directly would violate it.

Because concentration in this section enters as a regressor alongside a pay outcome, it is measured with employment rather than wage-bill shares, so that the regressor does not contain the outcome mechanically. The estimating system is

$$\ln(HHI_{m,t}) = \gamma \ln(HHI_{k,t}^{-l}) + \delta_m + \delta_t + \eta_{m,t}, \tag{8}$$

$$\overline{\psi}_{m,t} = \beta \ln(\widehat{HHI_{m,t}}) + \delta_m + \delta_t + \varepsilon_{m,t}, \tag{9}$$

where δ_m are market fixed effects and δ_t year fixed effects. Table 11 also reports specifications with alternative sets of fixed effects. The estimation sample covers the 190 private-sector markets formed by 19 industries and the 10 provinces over 2001–2019. Observations are weighted by 2007 market employment, and standard errors are clustered by market. The coefficient β is the elasticity of the market-level wage premium with respect to local concentration, identified from within-market changes in concentration driven by industry-wide forces.

6.2 Results

Table 11 reports the IV estimates under eight sets of fixed effects, and Appendix Table A1 the first stage. In the specification with market and year fixed effects (column 8), the elasticity is -0.028 (s.e. 0.010): a 10

percent increase in local concentration lowers the market-level wage premium by about 0.28 percent. With market fixed effects only (column 7), the estimate is -0.020 and not statistically significant. Specifications that rely on cross-sectional variation give mixed results: the estimate is positive with industry fixed effects (columns 2 and 4), close to zero with year or province fixed effects (columns 1, 3 and 5), and negative with industry and province fixed effects (-0.046, column 6). Comparisons across markets therefore mix the effect of concentration with persistent differences in industrial structure, which the market fixed effects remove.

The first stage is strong when the instrument exploits differences across markets, with cluster-robust F-statistics between 23 and about 1,490 in columns 1 to 6. Once market fixed effects absorb these differences, the identifying variation is limited to within-market changes in the leave-one-out measure, and the cluster-robust F-statistic falls to about 4 (columns 7 and 8). The market and year estimate should be read together with column 6, where the first stage remains strong ($F \approx 28$) and the elasticity is negative and significant at the 5 percent level (-0.046, s.e. 0.021).

The Canadian estimates are of the same order of magnitude as the elasticity of about -0.03 reported by Rinz (2022) for U.S. earnings under a comparable design. The two are complementary: Rinz measures the response of observed earnings, while the AKM-based outcome isolates the firm-pay component of earnings. Because the connected set includes public employers, private-sector firm effects are also measured against a broader set of opportunities, which may contribute to the somewhat smaller estimate in the preferred specification. Overall, the estimates indicate that higher local concentration lowers the firm component of pay.

Table 11: IV Estimates: Market-Level Wage Premium and Employment-Based Concentration

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\ln(HHI_{m,t})$	−0.0043	0.0372**	−0.0046	0.0412***	−0.0048	−0.0458*	−0.0196	−0.0278**
	(0.0078)	(0.0128)	(0.0081)	(0.0100)	(0.0081)	(0.0214)	(0.0184)	(0.0096)
Year FE	Yes	No	No	Yes	Yes	No	No	Yes
Industry FE	No	Yes	No	Yes	No	Yes	No	No
Province FE	No	No	Yes	No	Yes	Yes	No	No
Market FE	No	No	No	No	No	No	Yes	Yes
First-stage F (cluster-robust)	193	23	1,491	24	1,488	28	4.4	4.2
Observations	3,610	3,610	3,610	3,610	3,610	3,610	3,610	3,610
R^2	0.090	0.762	0.021	0.850	0.119	0.760	0.804	0.902

Notes: The dependent variable is the market-level wage premium $\overline{\psi}_{m,t}$, the employment-weighted mean of AKM firm effects (in log points) of private-sector firms in market m and year t. The regressor is the log employment-based HHI, instrumented with the leave-one-out measure of Equation 7. A market is a 2-digit NAICS industry in a province. All specifications use 2007 market-employment weights. Standard errors clustered by market in parentheses. With a single instrument, the cluster-robust (Kleibergen–Paap) first-stage F-statistic equals the squared t-statistic of the first-stage coefficient in Appendix Table A1. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

7 Fiscal Policy under Employer Market Power

The previous sections show that local concentration rises sharply in downturns and that higher concentration lowers the firm component of pay. This section asks how fiscal policy interacts with employer market power. I extend a two-agent New Keynesian model with an endogenous wage markdown and study the response to a government spending shock.

7.1 Model

The model builds on the TANK framework of Cantore and Freund (2021), with two types of households: workers, who earn labour income and save in bonds subject to a portfolio adjustment cost, and capitalists, who own the firms and receive their profits. I add monopsony power in the labour market, with a wage markdown that varies over the business cycle as in Bredemeier et al. (2026). Government spending is financed by lump-sum taxes and public debt.

7.1.1 Households

The economy is populated by a unit mass of households: a share λ of workers and a share $1-\lambda$ of capitalists. Workers consume C_t^W, supply N_t^W hours and hold real bonds B_t^W. Their preferences are

$$E_0 \sum_{t=0}^{\infty} \beta^t \left[\ln C_t^W - \frac{(N_t^W)^{1+\varphi}}{1+\varphi} \right], \tag{10}$$

and their budget constraint is

$$C_t^W + B_t^W + \frac{\psi^W}{2} \left(B_t^W - \bar{B}^W \right)^2 = W_t N_t^W + \frac{R_{t-1}}{\Pi_t} B_{t-1}^W - T_t, \tag{11}$$

where W_t is the real wage, R_t the gross nominal interest rate, Π_t gross inflation, T_t real lump-sum taxes and $\psi^W \geq 0$ the portfolio adjustment cost. Capitalists have preferences $E_0 \sum_t \beta^t \ln C_t^C$ and budget constraint

$$C_t^C + B_t^C = \frac{D_t}{1-\lambda} + \frac{R_{t-1}}{\Pi_t} B_{t-1}^C - T_t, \tag{12}$$

where D_t are aggregate profits. The first-order conditions are

$$\frac{1}{C_t^C} = \beta E_t \left[\frac{R_t}{\Pi_{t+1}} \frac{1}{C_{t+1}^C} \right], \tag{13}$$

$$\frac{1 + \psi^W \left(B_t^W - \bar{B}^W \right)}{C_t^W} = \beta E_t \left[\frac{R_t}{\Pi_{t+1}} \frac{1}{C_{t+1}^W} \right], \tag{14}$$

$$(N_t^W)^{\varphi} = \frac{W_t}{C_t^W}. \tag{15}$$

Aggregate consumption and hours are $C_t = \lambda C_t^W + (1-\lambda) C_t^C$ and $N_t = \lambda N_t^W$.

7.1.2 Firms and Monopsony Power

Firms produce with the technology $Y_t = N_t$ and sell differentiated goods with elasticity of substitution ε. Prices are set as in Calvo, with a probability θ that a firm cannot reset its price in a given quarter. In the labour market, firms have monopsony power: they pay a wage below the marginal revenue product of labour, $W_t = \mu_t \, \mathrm{MRPL}_t$, where MRPL_t equals real marginal cost because one hour produces one unit of output. The ratio $\mu_t \in (0,1]$ is the share of the marginal revenue product paid as wages: a higher μ_t means a narrower markdown. It depends on the elasticity of labour supply to the firm, η_t, through

$$\mu_t = \frac{\eta_t}{1+\eta_t}, \qquad \mu_{ss} = \frac{\eta_{ss}}{1+\eta_{ss}}. \tag{16}$$

This elasticity is linked to market structure: in the oligopsony model of Berger et al. (2022), the inverse of the effective elasticity is a payroll-weighted combination of the substitutability of workers across firms within a market and across markets, with weights given by the firm's share of the market wage bill, so that more concentrated markets have lower elasticities and wider markdowns.

Following Bredemeier et al. (2026), the elasticity varies with labour market conditions. In log deviations from the steady state (lower-case letters),

$$\hat{\eta}_t = \Gamma \left(w_t - c_t^W + n_t^W \right), \qquad \hat{\mu}_t = (1-\mu_{ss}) \, \hat{\eta}_t, \tag{17}$$

where $\Gamma > 0$ measures the sensitivity of the elasticity. By the labour supply condition (15), $w_t - c_t^W = \varphi n_t^W$, so the elasticity rises with hours: when the labour market tightens, firms compete harder for workers and the markdown narrows. Real marginal cost satisfies $mc_t = w_t - \hat{\mu}_t$, and inflation follows the New Keynesian Phillips curve

$$\pi_t = \beta E_t \pi_{t+1} + \kappa \, mc_t, \qquad \kappa = \frac{(1-\theta)(1-\beta\theta)}{\theta}. \tag{18}$$

Profits, which accrue to capitalists, move inversely with the real wage; in log deviations I use the approximation $d_t = -w_t$.

7.1.3 Government and Monetary Policy

The government finances spending G_t with lump-sum taxes and one-period debt B_t:

$$B_t = \frac{R_{t-1}}{\Pi_t} B_{t-1} + G_t - T_t. \tag{19}$$

Spending follows $g_t = \rho_g\, g_{t-1} + \varepsilon_t^g$, and taxes follow the rule $t_t = \phi_t\, t_{t-1} + \phi_b\, b_{t-1} + \phi_g\, g_t$. Bonds are held by both types of households, $B_t = \lambda B_t^W + (1-\lambda) B_t^C$. The central bank sets the nominal rate according to

$$r_t^n = \phi_R\, r_{t-1}^n + (1-\phi_R)\,\phi_\pi\,\pi_t + \varepsilon_t^m, \tag{20}$$

where r_t^n is the log deviation of R_t and ε_t^m a monetary policy shock, set to zero in the experiments below.

7.2 Calibration

The calibration draws on Canadian sources and is summarized in Table 12. The discount factor is $\beta = 0.99$ and the inverse Frisch elasticity $\varphi = 1.5$, consistent with the Frisch elasticities of about 0.5 to 0.75 reviewed by Chetty et al. (2011, 2013). Workers represent a share $\lambda = 0.75$ of households. The elasticity of substitution between goods is $\varepsilon = 6$, a 20 percent price markup (Galí, 2015), and the Calvo parameter $\theta = 0.71$ matches the average price duration of about 3.5 quarters in ToTEM III, the Bank of Canada's projection model (Corrigan et al., 2021). The Taylor rule has $\phi_\pi = 1.5$ and smoothing $\phi_R = 0.85$ (Dib, 2003; Corrigan et al., 2021). Government spending persistence is $\rho_g = 0.85$ (Bouakez and Rebei, 2007); taxes respond to debt with $\phi_b = 0.33$ and to spending with $\phi_g = 0.10$, without persistence ($\phi_t = 0$). On the labour market, the steady-state wage-to-MRPL ratio is $\mu_{ss} = 0.75$, which implies a firm-level labour supply elasticity of $\eta_{ss} = 3$. This is a moderate markdown: Yeh et al. (2022) estimate that U.S. manufacturing workers receive about 65 cents per dollar of marginal revenue product, and the elasticities estimated by Webber (2015) imply even lower ratios. The sensitivity of the elasticity is $\Gamma = 0.78$ (Bredemeier et al., 2026), and the portfolio adjustment cost is $\psi^W = 0.0742$ (Cantore and Freund, 2021).

7.3 Results

Figure 8 compares the responses to a government spending shock in the baseline TANK model (black lines) and in the model with cyclical monopsony (red dashed lines). Spending rises by 1 percent on impact and decays with persistence ρ_g. Hours rise by about 0.3 percent on impact and, because $Y_t = N_t$, output moves one-for-one with hours. Aggregate consumption falls by about 0.7 percent on impact and recovers gradually: higher spending crowds out private consumption. Inflation rises by about 0.3 percentage points on impact. Hours, output and consumption respond almost identically in the two models, so cyclical monopsony barely changes the aggregate effects of the shock. Inflation, however, is more persistent with monopsony, because the real wage rises more and keeps marginal costs higher for longer.

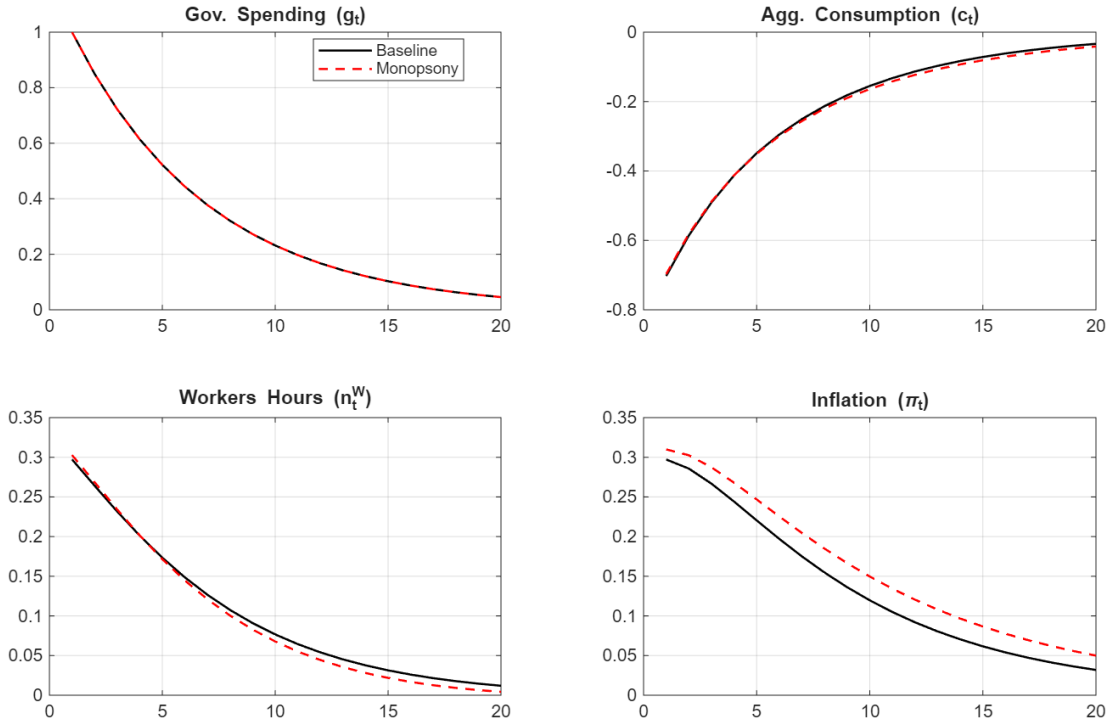
Figure 9 shows the mechanism. With monopsony, the real wage rises more strongly and stays elevated longer (a peak of about 0.30 percent, against 0.18 percent in the baseline), and profits fall more deeply and recover more slowly. As hours rise, the firm-level labour supply elasticity increases and the wage-to-MRPL ratio μ_t rises by about 0.15 percent on impact: firms pay a larger share of the marginal revenue product, so the markdown narrows. Because the real wage rises by more than μ_t, real marginal cost still increases. In line with Bredemeier et al. (2026), a fiscal expansion temporarily weakens firms' labour market power and shifts

Table 12: Calibration

Parameter	Value	Description	Source or target
Households			
β	0.99	Quarterly discount factor	Standard quarterly value
φ	1.5	Inverse Frisch elasticity	Chetty et al. (2011, 2013)
λ	0.75	Share of worker households	Households without profit income
ψ^W	0.0742	Portfolio adjustment cost	Cantore and Freund (2021)
Goods market			
ε	6	Elasticity of substitution between goods	20% price markup (Galí, 2015)
θ	0.71	Calvo probability	Price duration $\approx$ 3.5 quarters (ToTEM III)
Labour market			
μ_{ss}	0.75	Steady-state wage-to-MRPL ratio	Moderate markdown (Yeh et al., 2022)
η_{ss}	3.0	Firm-level labour supply elasticity	$\eta_{ss} = \mu_{ss}/(1-\mu_{ss})$
Γ	0.78	Sensitivity of the elasticity	Bredemeier et al. (2026)
Monetary policy			
ϕ_π	1.5	Response to inflation	Dib (2003); ToTEM III
ϕ_R	0.85	Interest rate smoothing	ToTEM III
Fiscal policy			
ρ_g	0.85	Persistence of government spending	Bouakez and Rebei (2007)
ϕ_t	0.0	Persistence of taxes	Taxes adjust immediately
ϕ_b	0.33	Tax response to debt	Debt stabilization
ϕ_g	0.10	Tax response to government spending	Partial tax financing on impact

Notes: Quarterly calibration. ToTEM III refers to Corrigan et al. (2021).

Figure 8: Aggregate Responses to a Government Spending Shock

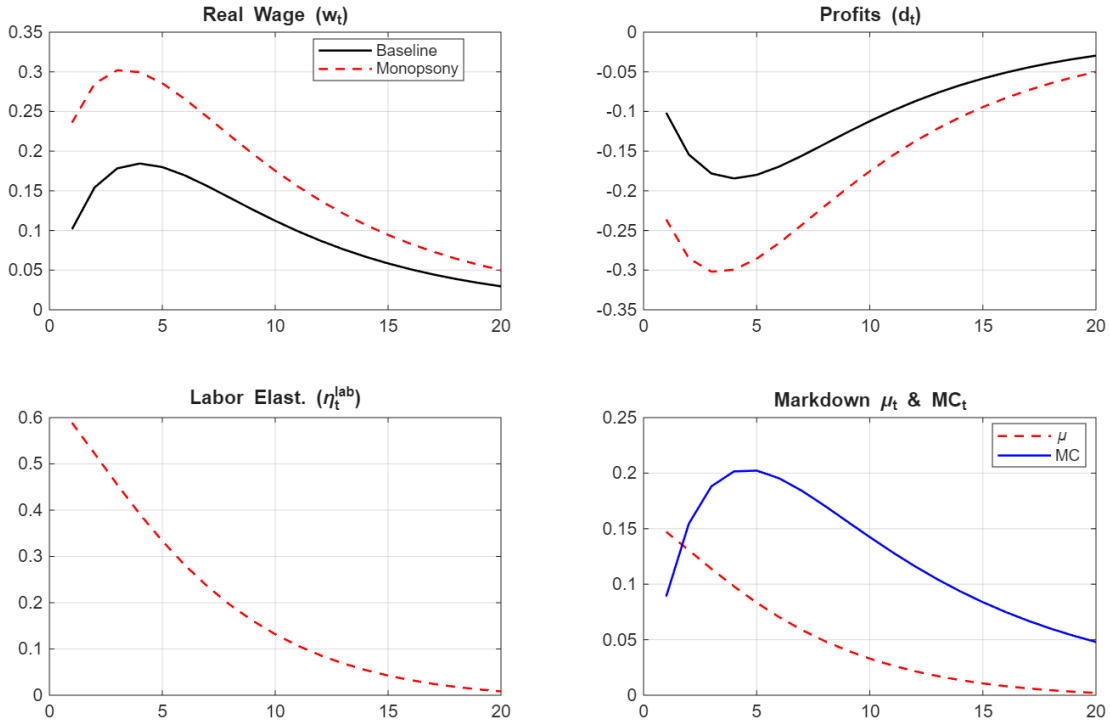


Notes: Impulse responses to a 1 percent government spending shock, in percent deviations from the steady state (inflation in percentage points); the horizontal axis is in quarters. Black solid lines: baseline TANK model; red dashed lines: model with cyclical monopsony.

Source: Model simulations.

surplus from profits to wages.

Figure 9: Wages, Profits and the Markdown After a Government Spending Shock



Notes: Impulse responses to a 1 percent government spending shock, in percent deviations from the steady state; the horizontal axis is in quarters. Top: real wage w_t and profits d_t in the baseline (black) and monopsony (red dashed) models. Bottom: firm-level labour supply elasticity $\hat{\eta}_t$, and the wage-to-MRPL ratio $\hat{\mu}_t$ (red dashed) with real marginal cost mc_t (blue) in the monopsony model.

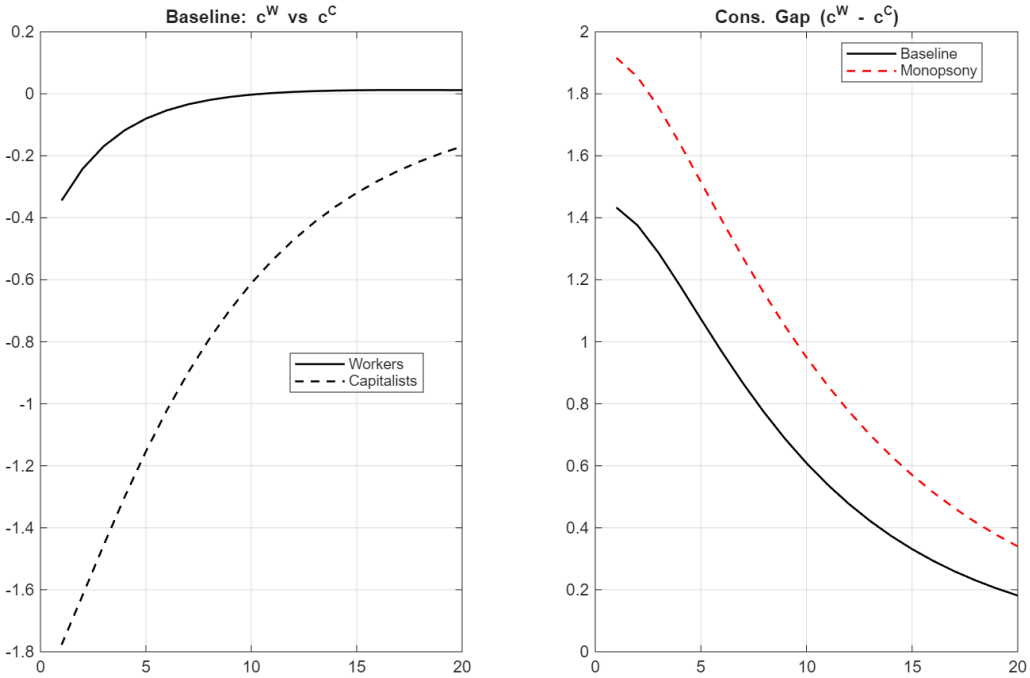
Source: Model simulations.

Figure 10 shows the distributional consequences. Capitalists, whose income depends on profits, cut consumption sharply, while workers' consumption falls much less (panel a, left). The consumption gap between workers and capitalists, $c_t^W - c_t^C$, therefore moves in workers' favour, by about 1.45 percent on impact in the baseline and about 1.9 percent with monopsony (panel a, right). The ratio of labour income to profits, $s_t = w_t + n_t - d_t$, rises by about 0.5 percent on impact in the baseline and by about 0.8 percent with monopsony, and the gap persists for several years (panel b). Fiscal policy, by narrowing markdowns during expansions, therefore redistributes more strongly from profits to wages in an economy with employer market power.

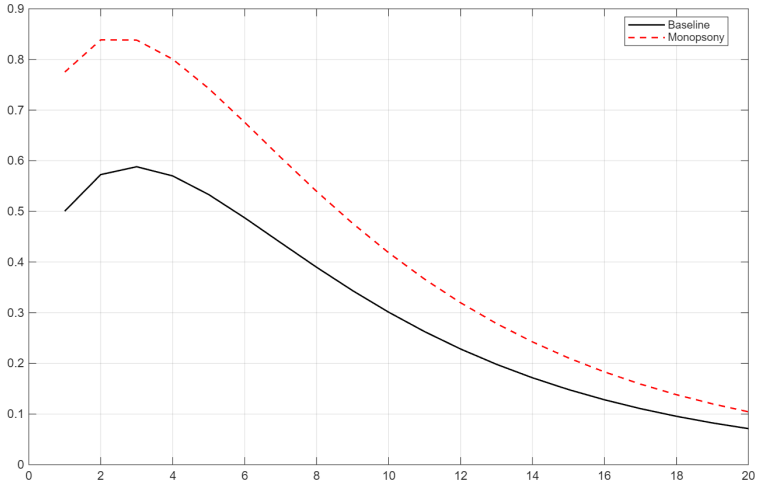
8 Conclusion

This paper documents how labour market concentration evolved in Canada between 2001 and 2019, with particular attention to the Great Recession and to the difference between national and local measures. Using wage-bill concentration measures built from administrative data, it shows that national and aggregate local concentration do not share the same time profile: national concentration declined gradually, while local

Figure 10: Distributional Effects of a Government Spending Shock



(a) Consumption of workers and capitalists (baseline, left) and consumption gap $c_t^W - c_t^C$ (right)



(b) Ratio of labour income to profits, $s_t = w_t + n_t - d_t$

Notes: Impulse responses to a 1 percent government spending shock, in percent deviations from the steady state; the horizontal axis is in quarters. Black solid lines: baseline TANK model; red dashed lines: model with cyclical monopsony.

Source: Model simulations.

concentration rose sharply during the Great Recession before resuming its decline.

The long-run decline reflects several distinct forces. Economy-wide, Public Administration accounts for 54 percent of the within-province decline, consistent with the federal restructuring after Budget 2012, and reallocation of the wage bill toward this sector drove most of the recession-era rise. In the private sector, the long-run decline came from falling concentration within industries, reinforced by a shift of the wage bill toward less concentrated industries. Continuing firms account for this deconcentration, while entry and exit offset part of it. During the Great Recession, private-sector concentration rose mainly within industries, and entry and exit added to the increase. Macroeconomic shocks therefore do not simply interrupt existing trends; they temporarily reshape the competitive structure of local labour markets.

The paper makes three further points. First, local concentration is much more responsive to recessions than national concentration, so national measures understate the cyclical sensitivity of employer power. Second, higher local concentration lowers the firm component of pay, with an elasticity of the market-level wage premium of about -0.03. Third, in a TANK model with a cyclical markdown, government spending narrows markdowns and shifts income from profits to wages, more strongly than in a model without cyclical monopsony. Understanding labour market power thus requires attention to local market structure, sectoral reallocation, firm dynamics, the role of public employers and the stance of fiscal policy.

References

John M. Abowd, Francis Kramarz, and David N. Margolis. High wage workers and high wage firms. *Econometrica*, 67(2):251–333, 1999.

David Autor, David Dorn, Lawrence F. Katz, Christina Patterson, and John Van Reenen. The fall of the labor share and the rise of superstar firms. *Quarterly Journal of Economics*, 135(2):645–709, 2020.

José Azar, Ioana Marinescu, and Marshall Steinbaum. Labor market concentration. *Journal of Human Resources*, 57(S):S167–S199, 2022.

Martin Neil Baily, Charles Hulten, and David Campbell. Productivity dynamics in manufacturing plants. *Brookings Papers on Economic Activity: Microeconomics*, 1992:187–267, 1992.

David Berger, Kyle Herkenhoff, and Simon Mongey. Labor market power. *American Economic Review*, 112 (4):1147–1193, 2022.

Hafedh Bouakez and Nooman Rebei. Why does private consumption rise after a government spending shock? *Canadian Journal of Economics*, 40(3):954–979, 2007.

Christian Bredemeier, Babette Jansen, and Roland Winkler. Labour market power and the effects of fiscal policy. *Economic Journal*, 2026. Forthcoming, doi:10.1093/ej/ueag023.

Cristiano Cantore and Lukas B. Freund. Workers, capitalists, and the government: Fiscal policy and income (re)distribution. *Journal of Monetary Economics*, 119:58–74, 2021.

David Card, Jörg Heining, and Patrick Kline. Workplace heterogeneity and the rise of West German wage inequality. *Quarterly Journal of Economics*, 128(3):967–1015, 2013.

Raj Chetty, Adam Guren, Day Manoli, and Andrea Weber. Are micro and macro labor supply elasticities consistent? A review of evidence on the intensive and extensive margins. *American Economic Review*, 101 (3):471–475, 2011.

Raj Chetty, Adam Guren, Day Manoli, and Andrea Weber. Does indivisible labor explain the difference between micro and macro elasticities? A meta-analysis of extensive margin elasticities. *NBER Macroeconomics Annual*, 27:1–56, 2013.

City of Ottawa. 2013 annual development report. Technical report, Planning and Growth Management Department, Research and Forecasting Unit, Ottawa, 2014.

Patrick Corrigan, Hugo Desgagnés, José Dorich, Vadym Lepetyuk, Wataru Miyamoto, and Yang Zhang. ToTEM III: The Bank of Canada's main DSGE model for projection and policy analysis. Technical Report 119, Bank of Canada, 2021.

Council of Economic Advisers. Labor market monopsony: Trends, consequences, and policy responses. Issue Brief, The White House, 2016.

Ali Dib. An estimated Canadian DSGE model with nominal and real rigidities. *Canadian Journal of Economics*, 36(4):949–972, 2003.

Guy Arnold Djolaud. *The Overlooked Role of the Public Sector in Labour Market Power*. Chapter 1 of *Three Essays on Labour Market Power*, PhD thesis, McGill University, 2026.

Benoît Dostie, Jiang Li, David Card, and Daniel Parent. Employer policies and the immigrant–native earnings gap. *Journal of Econometrics*, 233(2):544–567, 2023.

Lucia Foster, John C. Haltiwanger, and C. J. Krizan. Aggregate productivity growth: Lessons from microeconomic evidence. In *New Developments in Productivity Analysis*, pages 303–372. University of Chicago Press, 2001.

Jordi Galí. *Monetary Policy, Inflation, and the Business Cycle: An Introduction to the New Keynesian Framework and Its Applications*. Princeton University Press, second edition, 2015.

Simen Gaure. OLS with multiple high dimensional category variables. *Computational Statistics & Data Analysis*, 66:8–18, 2013.

Elizabeth Weber Handwerker and Matthew Dey. Measuring the relationship between employer concentration and wages. *Monthly Labor Review*, 2021.

Brad Hershbein, Claudia Macaluso, and Chen Yeh. Concentration in US local labor markets: Evidence from vacancy and employment data. Working paper, Federal Reserve Bank of Chicago, 2020.

Alan Manning and Barbara Petrongolo. Monopsony in local labour markets. *Oxford Open Economics*, 3 (Supplement 1):i951–i958, 2024.

Ioana Marinescu and Roland Rathelot. Mismatch unemployment and the geography of job search. *American Economic Journal: Macroeconomics*, 10(3):42–70, 2018.

Kevin Rinz. Labor market concentration, earnings, and inequality. *Journal of Human Resources*, 57(S): S251–S283, 2022.

Esteban Rossi-Hansberg, Pierre-Daniel Sarte, and Nicholas Trachter. Diverging trends in national and local concentration. *NBER Macroeconomics Annual*, 35(1):115–150, 2021.

Treasury Board of Canada Secretariat. Harper government announces 10,980 public sector positions eliminated in past six months. News release, Ottawa, 16 November 2012, 2012. URL `https://www.canada.ca/en/news/archive/2012/11/harper-government-announces-10-980-public-sector-positions-eliminated-past-six-months.html`.

Douglas A. Webber. Firm market power and the earnings distribution. *Labour Economics*, 35:123–134, 2015.

Chen Yeh, Claudia Macaluso, and Brad Hershbein. Monopsony in the US labor market. *American Economic Review*, 112(7):2099–2138, 2022.

A Additional Tables

Table A1: First-Stage Estimates: Leave-One-Out Concentration and Local HHI

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\ln(HHI_{k,t}^{-l})$	1.042	−1.017	1.004	−1.046	1.003	0.408	0.080	0.080
	(0.075)	(0.212)	(0.026)	(0.212)	(0.026)	(0.077)	(0.038)	(0.039)
Year FE	Yes	No	No	Yes	Yes	No	No	Yes
Industry FE	No	Yes	No	Yes	No	Yes	No	No
Province FE	No	No	Yes	No	Yes	Yes	No	No
Market FE	No	No	No	No	No	No	Yes	Yes
F-statistic (cluster-robust)	193	23	1,491	24	1,488	28	4.4	4.2
R^2	0.62	0.72	0.92	0.72	0.92	0.93	0.99	0.99
Observations	3,610	3,610	3,610	3,610	3,610	3,610	3,610	3,610

Notes: The dependent variable is the log employment-based HHI of market m. The instrument is the leave-one-out employment-weighted average HHI of the same industry in all other provinces (Equation 7), in logs. All specifications use 2007 market-employment weights. Standard errors clustered by market in parentheses. With a single instrument, the cluster-robust (Kleibergen–Paap) F-statistic equals the squared t-statistic of the first-stage coefficient.

Table A2: Decomposition of Economy-Wide ALC Change, Long Run (2001–2019)

Panel A: Overall decomposition

Total change	Within-province	Within-industry	Between-industry	Covariance
−0.076	−0.092 (119.6%)	−0.146 (160.2%)	0.067 (−73.5%)	−0.013 (14.0%)

Panel B: Industry contributions to the within-province effect

Industry (NAICS)	Within-industry	Between-industry	Covariance	Total	Share (%)
Agriculture, forestry & fishing (11)	0.001	0.000	0.000	0.001	−1.3
Mining & oil and gas extraction (21)	−0.003	0.006	−0.002	0.001	−0.8
Utilities (22)	−0.007	−0.009	0.000	−0.016	17.3
Construction (23)	−0.003	0.001	0.000	−0.002	1.8
Manufacturing (31–33)	0.005	−0.009	0.000	−0.005	5.3
Wholesale trade (41)	0.002	0.000	0.000	0.002	−1.6
Retail trade (44–45)	−0.001	0.000	0.000	−0.002	1.9
Transportation & warehousing (48–49)	−0.009	0.000	0.000	−0.008	8.9
Information & cultural industries (51)	−0.013	−0.014	0.005	−0.022	23.1
Finance & insurance (52)	0.001	0.006	0.000	0.008	−8.7
Real estate (53)	0.000	0.000	0.000	0.000	−0.4
Professional services (54)	0.000	0.001	0.000	0.001	−0.9
Management of companies (55)	0.004	−0.002	−0.001	0.001	−1.1
Administrative services (56)	−0.001	0.001	0.000	0.000	0.0
Educational services (61)	0.006	0.001	0.000	0.007	−7.0
Health care (62)	−0.012	−0.007	0.010	−0.009	10.1
Arts & entertainment (71)	−0.001	0.000	0.000	−0.001	0.8
Accommodation & food (72)	0.000	0.000	0.000	0.000	−0.2
Other services (81)	0.001	0.000	0.000	0.001	−1.2
Public administration (91)	−0.117	0.093	−0.025	−0.050	54.0
Within-province total				**−0.092**	**100.0**

Notes: All values are changes in aggregate local concentration (ALC) expressed as fractions of the 2001 ALC; for example, −0.075 is a 7.5 percent decline. Panel A splits the total change into the within-province effect and, within it, the within-industry, between-industry and covariance terms (Equations 4 and 5); the remainder of the total change is the between-province effect. Percentages in parentheses are shares of the immediate parent term. Panel B allocates the within-province effect across industries; the last numeric column is each industry's share of the within-province effect. Values are rounded to three decimals. All industries, including Public Administration (NAICS 91).

Table A3: Decomposition of Economy-Wide ALC Change, Great Recession (2007–2009)

Panel A: Overall decomposition

Total change	Within-province	Within-industry	Between-industry	Covariance
0.144	0.128 (88.9%)	0.040 (31.5%)	0.087 (67.9%)	0.001 (0.6%)

Panel B: Industry contributions to the within-province effect

Industry (NAICS)	Within-industry	Between-industry	Covariance	Total	Share (%)
Agriculture, forestry & fishing (11)	0.000	0.000	0.000	0.000	0.1
Mining & oil and gas extraction (21)	−0.001	−0.003	0.001	−0.004	−3.2
Utilities (22)	0.004	0.007	−0.001	0.010	7.7
Construction (23)	0.001	0.000	0.000	0.001	0.7
Manufacturing (31–33)	0.004	−0.003	0.000	0.002	1.4
Wholesale trade (41)	0.000	0.000	0.000	−0.001	−0.5
Retail trade (44–45)	0.000	0.000	0.000	0.001	0.5
Transportation & warehousing (48–49)	0.006	−0.001	0.000	0.004	3.4
Information & cultural industries (51)	0.007	−0.004	0.000	0.002	1.9
Finance & insurance (52)	0.005	−0.003	0.000	0.002	1.6
Real estate (53)	0.000	0.000	0.000	0.000	0.0
Professional services (54)	0.003	0.000	0.000	0.004	2.8
Management of companies (55)	0.001	0.000	0.000	0.001	1.1
Administrative services (56)	−0.002	0.000	0.000	−0.002	−1.5
Educational services (61)	0.004	−0.001	−0.002	0.002	1.3
Health care (62)	−0.001	0.002	0.001	0.002	1.4
Arts & entertainment (71)	0.000	0.000	0.000	0.001	0.6
Accommodation & food (72)	0.000	0.000	0.000	0.000	−0.2
Other services (81)	0.001	0.000	0.000	0.001	0.8
Public administration (91)	0.009	0.093	0.001	0.102	79.9
Within-province total				**0.128**	**100.0**

Notes: All values are changes in aggregate local concentration (ALC) expressed as fractions of the 2007 ALC; for example, −0.075 is a 7.5 percent decline. Panel A splits the total change into the within-province effect and, within it, the within-industry, between-industry and covariance terms (Equations 4 and 5); the remainder of the total change is the between-province effect. Percentages in parentheses are shares of the immediate parent term. Panel B allocates the within-province effect across industries; the last numeric column is each industry's share of the within-province effect. Values are rounded to three decimals. All industries, including Public Administration (NAICS 91).

Table A4: Within-Industry Contributions to Economy-Wide ALC Change, 2001–2019: Full Sample and Continuing Firms

Industry (NAICS)	Full sample	Continuing firms
Agriculture, forestry & fishing (11)	0.001	0.000
Mining & oil and gas extraction (21)	−0.003	0.002
Utilities (22)	−0.007	−0.004
Construction (23)	−0.003	−0.007
Manufacturing (31–33)	0.005	−0.006
Wholesale trade (41)	0.002	0.003
Retail trade (44–45)	−0.001	−0.006
Transportation & warehousing (48–49)	−0.009	−0.010
Information & cultural industries (51)	−0.013	−0.030
Finance & insurance (52)	0.001	−0.014
Real estate (53)	0.000	0.001
Professional services (54)	0.000	0.002
Management of companies (55)	0.004	0.005
Administrative services (56)	−0.001	−0.001
Educational services (61)	0.006	0.002
Health care (62)	−0.012	−0.016
Arts & entertainment (71)	−0.001	0.001
Accommodation & food (72)	0.000	0.000
Other services (81)	0.001	0.003
Public administration (91)	−0.117	−0.125
Total within-industry effect	**−0.146**	**−0.200**

Notes: Within-industry term of the decomposition in Equation 5, by industry, expressed as a fraction of the 2001 ALC. The first column is the full sample (Table A2); the second restricts the sample to firms present in both 2001 and 2019. All industries, including Public Administration (NAICS 91). Values are rounded to three decimals.

Table A5: Decomposition of Economy-Wide ALC Change, Great Recession (2007–2009): Continuing Firms

Panel A: Overall decomposition

Total change	Within-province	Within-industry	Between-industry	Covariance
0.118	0.096 (81.1%)	0.022 (23.1%)	0.042 (43.6%)	0.032 (33.3%)

Panel B: Industry contributions to the within-province effect

Industry (NAICS)	Within-industry	Between-industry	Covariance	Total	Share (%)
Agriculture, forestry & fishing (11)	0.000	0.000	0.000	0.000	0.3
Mining & oil and gas extraction (21)	−0.002	−0.002	0.002	−0.001	−1.4
Utilities (22)	0.005	0.007	−0.001	0.011	11.0
Construction (23)	0.001	0.000	0.000	0.001	1.5
Manufacturing (31–33)	0.007	−0.006	0.002	0.003	3.5
Wholesale trade (41)	−0.001	0.000	0.000	−0.001	−0.9
Retail trade (44–45)	−0.002	0.000	0.000	−0.002	−2.0
Transportation & warehousing (48–49)	0.011	0.000	0.000	0.011	11.6
Information & cultural industries (51)	−0.004	−0.004	0.001	−0.007	−7.5
Finance & insurance (52)	0.002	−0.001	0.000	0.001	1.2
Real estate (53)	0.000	0.000	0.000	0.000	0.5
Professional services (54)	0.007	0.002	0.002	0.011	11.5
Management of companies (55)	−0.001	0.000	0.000	−0.002	−1.7
Administrative services (56)	−0.004	−0.001	0.001	−0.005	−4.9
Educational services (61)	0.005	−0.002	−0.002	0.002	1.7
Health care (62)	−0.007	0.000	0.003	−0.004	−4.3
Arts & entertainment (71)	0.000	0.000	0.000	0.000	0.1
Accommodation & food (72)	−0.001	0.000	0.000	−0.001	−0.8
Other services (81)	0.002	0.000	0.000	0.003	2.6
Public administration (91)	0.002	0.049	0.024	0.075	78.1
Within-province total				**0.096**	**100.0**

Notes: All values are changes in aggregate local concentration (ALC) expressed as fractions of the 2007 ALC; for example, −0.075 is a 7.5 percent decline. Panel A splits the total change into the within-province effect and, within it, the within-industry, between-industry and covariance terms (Equations 4 and 5); the remainder of the total change is the between-province effect. Percentages in parentheses are shares of the immediate parent term. Panel B allocates the within-province effect across industries; the last numeric column is each industry's share of the within-province effect. Values are rounded to three decimals. All industries, including Public Administration (NAICS 91). The sample is restricted to firms present in every year from 2007 to 2009.

Table A6: Firm Entry and Exit by Industry During the Great Recession (2007–2009)

Industry (NAICS)	Entry (%)	Exit (%)	Net (%)
Agriculture, forestry & fishing (11)	18.82	20.48	−1.66
Mining & oil and gas extraction (21)	24.09	24.38	−0.29
Utilities (22)	18.07	19.34	−1.27
Construction (23)	23.90	20.31	3.59
Manufacturing (31–33)	13.70	17.86	−4.16
Wholesale trade (41)	14.54	18.20	−3.66
Retail trade (44–45)	18.19	21.09	−2.90
Transportation & warehousing (48–49)	26.04	22.38	3.66
Information & cultural industries (51)	24.50	23.06	1.44
Finance & insurance (52)	20.31	21.82	−1.51
Real estate (53)	24.24	21.42	2.82
Professional services (54)	24.55	20.21	4.34
Management of companies (55)	22.26	23.97	−1.71
Administrative services (56)	22.74	21.74	1.00
Educational services (61)	20.01	18.67	1.34
Health care (62)	19.33	14.03	5.30
Arts & entertainment (71)	18.31	18.66	−0.35
Accommodation & food (72)	23.70	24.59	−0.89
Other services (81)	22.52	22.83	−0.31
Public administration (91)	6.44	5.74	0.70

Notes: Firm entry and exit rates over 2007–2009, in percent of the number of firms in each industry; net entry is the entry rate minus the exit rate.